Simplifying Surds

AQA · KS4 Year 10–11

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE

72 —— = (36 × 2)

= 36 × 2

= 6 2

Simplifying a surd: split off the largest square factor.

COMMON MISTAKES

Not using the LARGEST square factor — do it in one step, not several.

(a + b) ≠ a + b. The rule only works for multiplication.

3 12 = 3 × 2 3 = 6 3, not 3 12.

a × a = a, not (a^2) left unsimplified.

METHOD

1. Find the largest square factor of the number
2. Split: n = (square × rest)
3. Use (a × b) = a × b
4. Square-root the perfect square out
5. Answer: k m (m has no square factors)

WORKED EXAMPLE

Simplify 72.

Step 1 — Largest square factor: 36

Step 2 — 72 = (36 × 2) = 36 × 2 = 6 2

VISUAL AID

Factor tree: 72 36 × 2 36 = 6 answer 6 2. Common square numbers reference: 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.

1. (2 marks)
Simplify: a) 12 b) 18 c) 20 d) 50

2. (2 marks)
Simplify: a) 8 b) 32 c) 45 d) 75

3. (3 marks)
Simplify: a) 98 b) 200 c) 128 d) 300

4. (3 marks)
Simplify: a) (4×7) b) 3 12 c) 2 50 d) 5 8

5. (3 marks)
Simplify fully: a) 48 + 12 b) 75 - 27 c) 2 18 + 3 2

6. (4 marks)
Show: a) 80 = 4 5 b) 3 28 = 6 7 c) 180 = 6 5

7. (4 marks)
Square area = 72 cm². a) Exact side? b) Perimeter? c) Compare with square of side 3 2.

8. (5 marks)
a) Show (a^2b) = a b. b) Simplify $(50x^4y^2)$. c) Right triangle sides 18, 32 — exact hypotenuse?